

Lista 2.

- 4) Encuentra la función generadora de momentos de X_n para la caminata aleatoria simple, y usala para encontrar su media y su varianza.

Recordatorio

$$P(X_{n+1}=j | X_n=i) = \begin{cases} p & \text{si } j=i+1 \\ q & \text{si } j=i-1 \\ 0 & \text{en otro caso} \end{cases}$$

$$\begin{aligned} & \bullet M(t) = E(e^{tx}) \\ & \bullet P(X_n=k) = \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} (1-p)^{\frac{n-k}{2}} \\ & \bullet E[f(x)] = \sum_x f(x) P(X=x) \end{aligned}$$

Consideremos a $X_n = X_0 + W_1 + W_2 + \dots + W_n$ donde los W_i con $i=1, \dots, n$ son variables aleatorias idénticas a W e independientes donde $W = X_1$ variable aleatoria.

$$\begin{aligned} E[e^{tX_n}] &= E[e^{t(X_0 + W_1 + \dots + W_n)}] = E[e^{tX_0 + tW_1 + \dots + tW_n}] \\ &= E[e^{tX_0} \cdot e^{tW_1} \cdot \dots \cdot e^{tW_n}] \end{aligned}$$

Dado que son independientes $\Rightarrow$

Idénticas $\Rightarrow$

$$\begin{aligned} &= E[e^{tX_0}] \cdot E[e^{tW_1}] \cdot \dots \cdot E[e^{tW_n}] \\ &= E[e^{tX_0}] \cdot (E[e^{tW}])^n \end{aligned} \quad (1)$$

Donde tenemos que

$$\begin{aligned} E[e^{tW}] &= \sum_w e^{tw} P(W=w) = \sum_w e^{tw} f_W(w) \\ &= e^{t \cdot 1} \cdot p + e^{t \cdot (-1)} \cdot q \end{aligned} \quad M_1(t)$$

Debido que $f_W(w) = \begin{cases} p & \text{si } w=1 \\ q & \text{si } w=-1 \end{cases}$ donde $p+q=1$

De esta manera, $E[e^{tW}] = (pe^t + qe^{-t})$ así
regresando a (1) tenemos que

$$\begin{aligned} E[e^{tX_n}] &= E[e^{tX_0}] \cdot (E[e^{tW}])^n \\ &= E[e^{tX_0}] \cdot (pe^t + qe^{-t})^n \\ &= E[1] \cdot (pe^t + qe^{-t})^n \quad (\text{considerando } X_0 = 0) \\ &= (pe^t + qe^{-t})^n \end{aligned}$$

$$\therefore M(t) = (pe^t + qe^{-t})^n$$

$$M'(t) = n(pe^t + qe^{-t})^{n-1} \cdot (pe^t - qe^{-t})$$

$$M'(0) = n \underbrace{(p+q)}_1^{n-1} \cdot (p-q) = n(p-q)$$

$$\begin{aligned} M''(t) &= n(n-1)(pe^t + qe^{-t})^{n-2} (pe^t - qe^{-t}) \cdot (pe^t - qe^{-t}) + \\ &\quad n(pe^t + qe^{-t})^{n-1} (pe^t + qe^{-t}) \end{aligned}$$

$$= n(n-1)(pe^t + qe^{-t})^{n-2} (pe^t - qe^{-t})^2 + n(pe^t + qe^{-t})^n$$

$$= n(n-1)(pe^t + qe^{-t})^{n-2} (pe^t - qe^{-t})^2 + nM(t)$$

$$M''(0) = n(n-1)(p-q)^2 + n = n((n-1)(p-q)^2 + 1)$$

$$E[X] = n(p-q), \quad E[X^2] = n((n-1)(p-q)^2 + 1)$$

$$\text{Var}[X] = E[X^2] - E^2[X] = n(n-1)(p-q)^2 + n - n^2(p-q)^2$$

$$= n(p-q)^2 (n-1-n) + n = -n(p-q)^2 + n$$

$$= n(1 - (p-q)^2) = n((p+q)^2 - (p-q)^2) = n(4pq)$$

$$r + r^2 + \dots + r^K$$

$$r + r^2 + \dots + r^K(1-r) = r - r^{K+1}$$

$$r + r^2 + \dots + r^K = \frac{r - r^{K+1}}{1-r}$$

$$0 < e^t(1-p) < 1$$

$$\frac{e^t(1-p)}{1 - e^t(1-p)}$$

1) Usar la función generadora de momentos de la distribución geométrica para encontrar su valor esperado y su varianza.

$$\frac{A}{B} \Rightarrow \frac{A'B - AB'}{B^2}$$

Recordatorio

$$X \sim \text{geométrica}, P(X=k) = (1-p)^{k-1} \cdot p, \quad M_X(t) = E[e^{xt}]$$

$$\text{Var}[X] = E[X^2] - E^2[X]$$

$$\begin{aligned} M_X(t) &= E[e^{kt}] = \sum_{k=1}^{\infty} e^{kt} (1-p)^{k-1} \cdot p = \sum_{k=1}^{\infty} e^{kt} (1-p)^{k-1} \cdot p \cdot \frac{(1-p)}{1-p} \\ &= \frac{p}{1-p} \sum_{k=1}^{\infty} [e^t(1-p)]^k = \frac{p}{1-p} \left[\frac{(1-p)e^t}{1 - (1-p)e^t} \right] \\ &= \frac{pe^t}{1 - (1-p)e^t} \end{aligned}$$

De esta manera $M(t) = \frac{pe^t}{1 - (1-p)e^t}$

$$M'(t) = \frac{pe^t(1 - (1-p)e^t) - pe^t(-(1-p)e^t)}{(1 - (1-p)e^t)^2}$$

$$= \frac{pe^t(1 - (1-p)e^t + (1-p)e^t)}{(1 - (1-p)e^t)^2}$$

$$= \frac{pe^t pe^t}{(1 - (1-p)e^t)^2}$$

$$M'(t) = \frac{p}{(1 - (1-p))^2} = \frac{p}{p^2} = \frac{1}{p}$$

$$M''(t) = \frac{pe^t (1-(1-p)e^t)^2 - 2(1-(1-p)e^t)(-(1-p)e^t) \cdot pe^t}{(1-(1-p)e^t)^4}$$

$$= \frac{pe^t (1-(1-p)e^t) + 2(1-p)e^t pe^t}{(1-(1-p)e^t)^3}$$

$$= \frac{pe^t (1-(1-p)e^t + 2(1-p)e^t)}{(1-(1-p)e^t)^3} = \frac{pe^t (1 + (1-p)e^t)}{(1-(1-p)e^t)^3}$$

$$M''(0) = \frac{p(1+(1-p))}{(1-(1-p))^3} = \frac{p(2-p)}{p^3} = \frac{2-p}{p^2}$$

De esta manera $E[X] = \frac{1}{p}$, $E[X^2] = \frac{2-p}{p^2}$ así

$$\text{Var}(X) = E[X^2] - E^2[X]$$

$$= \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}$$